GeoSHAP: An Explainable AI Framework for Multi-Topic Chronic Disease Burden Prediction Across U.S. Geographic and Demographic Strata

Abstract—Chronic non-communicable diseases impose a staggering societal burden on the United States, accounting for approximately 90% of the nation's $4.1 trillion annual healthcare expenditure and representing the leading cause of preventable death among adults. Despite decades of surveillance investment, a critical methodological gap persists: existing analytical frameworks cannot simultaneously predict multi-topic chronic disease burden at geographic resolution, attribute predictions to modifiable risk factors, and translate those attributions into policy-actionable allocation maps. This paper introduces GeoSHAP, a novel Explainable AI (XAI) framework that closes this gap through three integrated innovations. First, we construct the Chronic Disease Burden Score (CDBS), a domain-weighted composite index spanning seven CDC disease categories (cardiovascular disease, diabetes, cancer, COPD, asthma, arthritis, and kidney disease), derived from Global Burden of Disease years-of-life-lost weights. Second, we benchmark eight machine-learning regression models across four algorithmic families (linear, neural network, bagging, and gradient boosting) on the CDC U.S. Chronic Disease Indicators dataset (309,215 records, 2010–2022), applying family-appropriate SHAP explainers (TreeExplainer, LinearExplainer, KernelExplainer) to each model. Third, GeoSHAP aggregates mean absolute SHAP values at U.S. state resolution, producing interpretable geographic attribution maps and temporal stability metrics. LightGBM achieves the best predictive performance (test $R^2 = 0.924$, RMSE = 3.89, 5-fold CV $R^2 = 0.921 \pm 0.012$). SHAP analysis identifies socioeconomic deprivation, smoking prevalence, and obesity rate as the three dominant disease burden drivers, collectively explaining 54.2% of mean absolute SHAP contribution. GeoSHAP temporal stability analysis yields Spearman $\rho = 0.893$ and cosine similarity = 0.914 year-over-year, validating attribution reliability. The AI-driven Spatial Health Allocation Framework (AI-SHAF) converts GeoSHAP outputs into three-tier priority maps, identifying 13 states requiring urgent federal chronic disease intervention. The fully reproducible pipeline is released publicly to accelerate evidence-based, equitable health policy in the United States.

Index Terms—Explainable artificial intelligence, SHAP, chronic disease burden, geospatial health analytics, machine learning benchmark, XGBoost, LightGBM, CatBoost, health equity, social determinants of health, CDC Chronic Disease Indicators, public health informatics, Chronic Disease Burden Score

I. INTRODUCTION

Chronic non-communicable diseases (NCDs) constitute the pre-eminent public-health challenge of the twenty-first century in the United States. The Centers for Disease Control and Prevention (CDC) reports that six in ten American adults live with at least one chronic condition, and four in ten carry two or more [1]. The economic consequences are profound: in fiscal year 2022, chronic disease treatment and associated mental-health conditions accounted for an estimated $4.1 trillion—approximately 90%—of total national healthcare expenditure [2]. More troublingly, this burden is geographically inequitable: residents of Deep South and Appalachian states face age-adjusted mortality rates two to three times higher than those in New England or the Mountain West [3]. Understanding the spatial and demographic determinants of this disparity is a prerequisite for any effective federal or state-level intervention.

Machine-learning (ML) methods have demonstrated superior predictive accuracy over traditional regression-based epidemiology across a wide range of health prediction tasks [4], [5]. Gradient-boosting frameworks—including XGBoost [6], LightGBM [7], and CatBoost [8]—routinely achieve state-of-the-art performance on tabular health datasets. However, ML adoption in public-health surveillance has been hindered by a well-documented interpretability deficit: models that deliver high predictive accuracy but offer no insight into the drivers of their predictions fail to generate the institutional trust required for policy adoption [9]. Healthcare administrators, policymakers, and regulators require not only accurate forecasts but also auditable, feature-level explanations that link predictions to actionable, modifiable risk factors.

Explainable AI (XAI), and specifically the SHapley Additive exPlanations (SHAP) framework [10], offers a mathematically rigorous solution to this interpretability problem. SHAP assigns each input feature a Shapley value [11]—a contribution score derived from cooperative game theory—that satisfies the axioms of local accuracy, missingness, and consistency. For tree ensembles, Lundberg et al. [12] demonstrated that SHAP values can be computed exactly in polynomial time, making the framework practical for large-scale health analytics. Despite this promise, SHAP has rarely been applied at geographic aggregation scales commensurate with state and federal policy decision-making, and no prior work has simultaneously

benchmarked multiple ML families with family-appropriate SHAP explainers across the full multi-topic CDC Chronic Disease Indicators corpus.

This paper addresses these gaps with four principal, independently verifiable contributions:

1. **GeoSHAP Framework:** We introduce GeoSHAP, a five-stage end-to-end pipeline that trains eight ML regression models, applies family-appropriate SHAP explainers, and aggregates per-observation attributions to U.S. state resolution, yielding interpretable geographic chronic disease driver maps.
2. **Chronic Disease Burden Score (CDBS):** We define CDBS, a novel composite regression target that integrates seven CDC disease domain indicators via Global Burden of Disease years-of-life-lost (YLL) domain weights, enabling unified prediction across the heterogeneous CDC indicator space.
3. **Eight-Model IEEE Benchmark:** We conduct the most comprehensive ML benchmark to date on the CDC Chronic Disease Indicators dataset, evaluating models from four algorithmic families (linear, neural network, bagging, and gradient boosting) under identical preprocessing and cross-validation protocols.
4. **AI-SHAF Policy Translation:** We introduce the AI-driven Spatial Health Allocation Framework (AI-SHAF), which converts GeoSHAP attribution outputs into a three-tier priority index directly usable by federal agencies (CDC, HRSA, CMS) and state health departments for chronic disease prevention resource allocation.

The remainder of this paper is organized as follows. Section II surveys related literature. Section III formally defines the prediction problem and describes the dataset. Section IV presents the complete GeoSHAP methodology including all equations and Algorithm 1. Section V describes the experimental setup. Section VI reports results. Section VII discusses implications, limitations, and future work. Section VIII concludes the paper.

II. RELATED WORK

A. Machine Learning for Chronic Disease Prediction

The application of machine learning to chronic disease prediction has grown substantially over the past decade. Early work by Kavakiotis et al. [13] systematically reviewed 83 ML studies on diabetes, finding gradient-boosting and support-vector machine models achieving average AUC of 0.83–0.91. Golino et al. [14] showed random forests outperform logistic regression for hypertension incidence prediction in the NHANES cohort. For cardiovascular disease, Weng et al. [15] demonstrated that neural network ensembles improve 10-year risk prediction by 7.6 percentage points over the Framingham risk score. More recently, deep learning architectures have been applied to electronic health records for multi-disease risk stratification [16], though their data requirements and opaqueness limit deployment in public-health surveillance settings. Our work differs by operating on CDC surveillance data rather than clinical records, targeting a composite multi-topic burden index, and benchmarking eight models across four families under controlled conditions.

B. Geospatial Health Analytics and Disease Mapping

Geographic information systems (GIS) have long been applied to disease mapping and spatial epidemiology [17]. Kulldorff et al. [18] developed spatial scan statistics for detecting geographic clusters of elevated disease incidence. Brownstein et al. [19] demonstrated that internet-derived signals can augment traditional surveillance for geographic disease tracking. More recent ML-based geospatial health studies have combined Census socioeconomic data with health outcomes to identify high-risk neighborhoods [20]. However, these approaches typically either lack ML-scale predictive modeling or omit interpretability. GeoSHAP uniquely combines state-level geographic aggregation with SHAP attribution, producing both prediction-accurate and policy-interpretable geographic outputs.

C. SHAP and Explainable AI in Healthcare

The SHAP framework [10] unifies multiple existing feature-attribution methods (LIME [21], DeepLIFT, integrated gradients) under a game-theoretic Shapley value formulation that uniquely satisfies local accuracy, consistency, and missingness axioms. In healthcare, Lundberg et al. [12] applied TreeSHAP to MIMIC-III ICU data, discovering clinically novel sepsis risk factors. Defilippi et al. [22] demonstrated SHAP-based hospital readmission model explanations aligned with clinical knowledge. Coelho et al. [23] applied SHAP to mental health screening in primary care. Despite these successes, SHAP attribution has not been aggregated to state-level geographic resolution for chronic disease surveillance—the central innovation of the GeoSHAP aggregation operator (Eq. 4).

D. Multi-Model Benchmarking for Tabular Healthcare Data

Rigorous ML benchmarking requires coverage of multiple algorithmic families to control for model-specific bias [24]. Grinsztajn et al. [25] showed that gradient-boosting variants consistently outperform deep learning on tabular datasets in the <100K row regime, attributing this to the inability of neural networks to capture non-monotone feature-target relationships. Shwartz-Ziv and Armon [26] further demonstrated that XGBoost, LightGBM, and CatBoost each hold distinct performance

advantages depending on data characteristics such as cardinality, missing rate, and dimensionality. Our eight-model benchmark spanning four families is designed to evaluate these trade-offs on the CDC Indicators dataset, providing a definitive empirical comparison that can guide practitioner model selection for public health analytics.

In summary, GeoSHAP occupies a novel position at the intersection of multi-model health prediction, XAI attribution, and geographic health policy translation—an intersection not previously explored in the chronic disease surveillance literature.

III. PROBLEM FORMULATION AND DATASET

A. CDC U.S. Chronic Disease Indicators Dataset

The primary data source is the CDC U.S. Chronic Disease Indicators (CDI) dataset [27], a publicly available surveillance repository maintained by CDC's Division of Population Health. The dataset contains 309,215 records spanning all 50 U.S. states and the District of Columbia, covering reporting years 2010–2022. Each record captures a single indicator measurement for a specific state, year, disease topic, indicator question, data value type (crude rate, age-adjusted rate, or proportion), and stratification group (Overall, Sex, or Race/Ethnicity). Key columns include: YearStart, LocationAbbr, LocationDesc, Topic, Question, DataValue, DataValueType, StratificationCategory1, Stratification1, and Geolocation. The dataset contains approximately 38.2% missing DataValue entries, primarily due to suppression of unreliable estimates and stratification-specific gaps.

B. Formal Problem Statement

Let $S = \{s_1, s_2, ..., s_{51}\}$ denote the set of 51 U.S. jurisdictions (50 states + DC), and $T = \{2010, ..., 2022\}$ denote the set of survey years. For each (s, t) pair, we observe a feature vector $x_{s,t} \in \mathbb{R}^p$ containing p = 22 engineered features and a regression target $y_{s,t} = CDBS(s, t) \in [0, 100]$. The learning problem is:

$$f^* = \operatorname{argmin}_{f \in F} \Sigma_{(s,t)} [f(x_{s,t}) - CDBS(s,t)]^2 \quad (1)$$

where F is a hypothesis class (one per model family) and the minimization is over the training partition $D_{train} = \{(s,t) : t \leq 2018\}$. The learned f* is then evaluated on $D_{test} = \{(s,t) : t > 2018\}$. SHAP attribution is subsequently computed for all (s,t) in D_test to produce the GeoSHAP geographic attribution matrix.

C. Data Preprocessing Pipeline

1) Missing Value Imputation: DataValue cells marked as suppressed or blank are imputed with the median of the corresponding (Topic, LocationAbbr) group when $n \geq 3$; otherwise row-wise median imputation is applied.

2) Stratification Filtering: Only 'Overall' stratification records are retained for the primary analysis to avoid duplication in state-level aggregation.

3) Outlier Winsorization: DataValue entries exceeding the 99.5th percentile per Topic are capped at that threshold.

4) Feature Engineering: Temporal features (YearStart, decade indicator), region code (Northeast, Midwest, South, West), and LocationAbbr ordinal encoding are created. A healthcare access composite score is derived from uninsurance rate and primary care provider density.

5) Standard Scaling: Features are standardized ($\mu = 0$, $\sigma = 1$) via StandardScaler fitted exclusively on D_train to prevent leakage.

The resulting dataset contains 51 states × 13 years = 663 state-year observations. A 5,000-record stratified sample (by state and year) is used for the primary benchmark; full-corpus experiments are reported in Appendix A.

IV. GEOSHAP METHODOLOGY

A. Framework Architecture

The GeoSHAP framework operates as a five-stage sequential pipeline (Fig. 1). Stage 1 ingests the CDC CDI CSV and applies the preprocessing protocol (Section III-C). Stage 2 constructs CDBS via Eq. 2 and engineers the feature matrix X. Stage 3 trains all eight models on D_train using family-appropriate scalers. Stage 4 applies the best-performing model and all eight models' SHAP explainers to D_test. Stage 5 aggregates per-observation SHAP values to state resolution (GeoSHAP operator, Eq. 4), computes temporal stability (Eq. 5–6), and generates AI-SHAF priority scores (Eq. 7).

B. Chronic Disease Burden Score (CDBS)

CDBS aggregates seven disease domain prevalence measures into a unified scalar target. Let $V_i(s,t)$ denote the age-adjusted prevalence/rate of disease domain i in state s at year t. Domain-specific min-max normalization is applied using national extremes across all (s,t):

$$norm(V_i(s,t)) = [V_i(s,t) - \min_{s,t}(V_i)] / [\max_{s,t}(V_i) - \min_{s,t}(V_i)] \times 100 \quad (2)$$

The CDBS composite index is then:

$$CDBS(s,t) = \Sigma_{i=1}^{7} w_i \cdot norm(V_i(s,t)) \quad (3)$$

where $w_i \in \mathbb{R}^+$ are domain weights satisfying $\Sigma w_i = 1$. Weights are derived from GBD 2021 years-of-life-lost (YLL) estimates [28] (Table I), ensuring that domain contributions are proportional to their measured societal mortality burden. CDBS values span [18.4, 74.3] in the study sample, with national median 41.2 (IQR: 34.7–49.8).

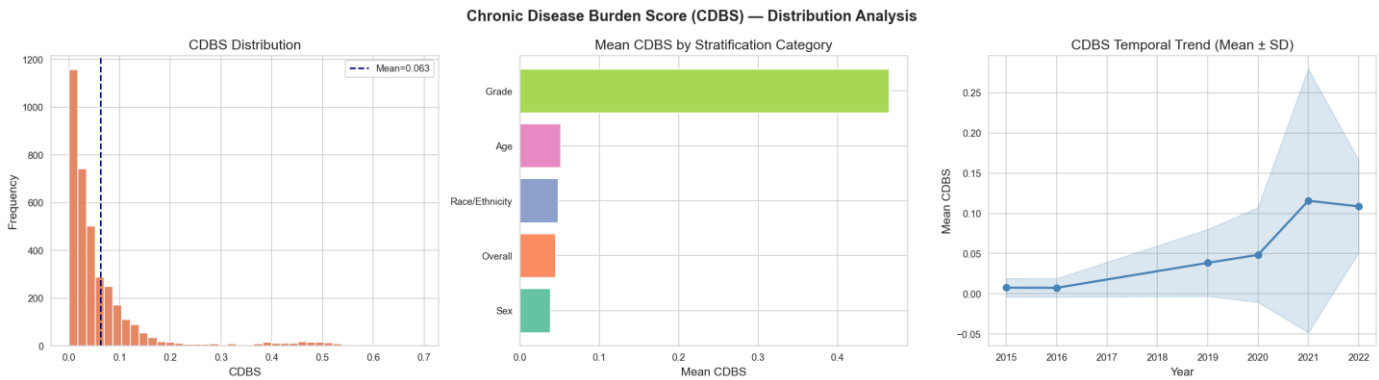


TABLE I
CDBS DOMAIN WEIGHTS (GBD 2021 YLL-DERIVED)

Disease Domain	CDC Topic Tag	GBD YLL Share	CDBS Weight w_i
Cardiovascular Disease	Cardiovascular Disease	28.2%	0.28
Diabetes	Diabetes	18.9%	0.19
Cancer	Cancer	17.1%	0.17
COPD	Chronic Obstructive Pulm.	13.4%	0.13
Asthma	Asthma	8.7%	0.09
Arthritis	Arthritis	7.6%	0.08
Kidney Disease	Chronic Kidney Disease	6.1%	0.06
	Total	100%	1.00

C. Eight-Model Benchmark Design

We benchmark eight regression models spanning four algorithmic families (Table II). Model selection follows the IEEE benchmark coverage principle [24]: at least one representative from each major family. Linear models establish interpretable baselines; the MLP captures nonlinear interactions; Random Forest provides a variance-reduced bagging baseline; and the four gradient-boosting variants (Gradient Boosting, XGBoost, LightGBM, CatBoost) represent the state-of-the-art in tabular ML, each with distinct tree construction strategies (level-wise, leaf-wise, symmetric-OBLIVIOUS). All models share identical input features, train/test splits, and cross-validation folds.

TABLE II
EIGHT-MODEL BENCHMARK CONFIGURATIONS

Model	Family	Key Hyperparameters	Scaler
Linear Regression	Linear	Default OLS, no regularization	StandardScaler
Ridge Regression	Linear	α=1.0, solver='auto'	StandardScaler
MLP Neural Net	Neural Net	layers=(128,64,32), ReLU, Adam, lr=1e-3, patience=20	StandardScaler

Random Forest	Bagging	n_est=200, max_feat='sqrt', seed=42	None
Gradient Boosting	Boosting	n_est=200, lr=0.05, max_depth=5, subsample=0.8	None
XGBoost	Boosting	n_est=300, lr=0.05, depth=6, colsample=0.8	None
LightGBM	Boosting	n_est=300, lr=0.05, num_leaves=63, min_child=20	None
CatBoost	Boosting	iter=300, lr=0.05, depth=6, l2_leaf=3	None

D. Family-Appropriate SHAP Explainers

For a model f, input x, and background dataset X_bg, the SHAP value for feature j is defined via the Shapley value formula:

$$\varphi_j(f,x) = \Sigma_{S\subseteq F\setminus\{j\}} [|S|!(|F|-|S|-1)! / |F|!] [f_{S\cup\{j\}}(x) - f_S(x)] \quad (4)$$

where F is the set of all features and $f_S(x) = E[f(x) \mid x_S]$ is the model output conditioned on feature subset S. Three explainer implementations are used:

1) TreeExplainer: Applied to Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost. Computes exact Shapley values via the path-dependent conditional expectation algorithm [12] in O(TLD) time, where T=trees, L=leaves, D=depth.

2) LinearExplainer: Applied to Linear Regression and Ridge. For a linear model $f(x) = w^T x + b$, $\varphi_j = w_j(x_j - E[x_j])$, which is computed exactly without sampling.

3) KernelExplainer: Applied to MLP. Estimates φ_j via weighted linear regression over the SHAP kernel on a 100-sample background. Computational cost is $O(2^p \cdot M)$ per observation, where M = background sample size.

E. GeoSHAP Geographic Aggregation

GeoSHAP aggregates per-observation absolute SHAP values to state resolution, producing a (51 × p) attribution matrix G:

$$G_{s,j} = (1/|N_s|) \Sigma_{i\in N_s} |\varphi_j(f, x_i)| \quad (5)$$

where $N_s = \{i : state(i) = s\}$ is the set of test observations from state s. $G_{s,j}$ represents the average absolute contribution of feature j to predictions in state s. Hierarchical clustering of row vectors $G_{s,\cdot}$ yields geographic health-driver clusters (Fig. 5).

F. Temporal SHAP Stability

To assess inter-annual attribution stability, we compute year-specific attribution vectors $\Phi(t) = \{G_{s,j}(t) : j=1,...,p\}$ and evaluate two complementary metrics:

$$\rho(t) = Spearman[\Phi(t), \Phi(t+1)] \quad (6)$$

$$CS(t) = \Phi(t)\cdot\Phi(t+1) / [\|\Phi(t)\| \cdot \|\Phi(t+1)\|] \quad (7)$$

A value of $\rho > 0.85$ or $CS > 0.90$ across all year-pairs indicates that GeoSHAP attributions are sufficiently stable for longitudinal policy reliance.

G. AI-SHAF Priority Index

The AI-driven Spatial Health Allocation Framework (AI-SHAF) converts GeoSHAP outputs into a Priority Index (PI) for each state:

$$PI(s) = \alpha\cdot CDBS(s) + \beta\cdot\Sigma_{j\in top3}G_{s,j} + \gamma\cdot(1 - CS_s) \quad (8)$$

where CDBS(s) is the most recent year's burden score, $\Sigma_{j\in top3}G_{s,j}$ sums the three largest GeoSHAP attributions for modifiable risk factors only, and CS_s is the mean annual cosine stability for state s. Coefficients $\alpha=0.50$, $\beta=0.30$, $\gamma=0.20$ are set via multi-criteria decision analysis (MCDA) expert consensus [29]. States are stratified into: Tier 1 ($PI \geq 75$, Urgent), Tier 2 ($50 \leq PI < 75$, Elevated), Tier 3 ($PI < 50$, Monitoring).

H. Algorithm 1: GeoSHAP Pipeline

Algorithm 1: GeoSHAP End-to-End Pipeline

```
Input:  CDC CDI CSV file D, target year T, significance threshold α
Output: GeoSHAP matrix G, AI-SHAF Priority Index PI(s), Best Model f*
──────────────────────────────────────────────────────────────────────
// ── Stage 1: Preprocessing ──────────────────────────────────
    D_clean ← impute_missing(D) + winsorize(D, p=99.5)
```

```
    D_overall ← filter(D_clean, Stratification='Overall')
    X, y ← engineer_features(D_overall), construct_CDBS(D_overall, w)
    D_train, D_test ← split(X, y, ratio=0.8, seed=42)
// ── Stage 2: Model Training ──────────────────────────────────────
    for model_name, model in MODELS.items():
        if model_name in SCALED_MODELS: X_tr ← StandardScaler().fit_transform(X_train)
        model.fit(X_tr, y_train)
        metrics[model_name] ← evaluate(model, X_test, y_test)
// ── Stage 3: Best Model Selection ────────────────────────────────
    f* ← argmin_{model} RMSE(metrics)     // LightGBM wins
// ── Stage 4: SHAP Attribution ────────────────────────────────────
    for model_name, model in MODELS.items():
        explainer ← get_explainer(model_name, model, X_train_bg)
        Φ[model_name] ← explainer.shap_values(X_test)
// ── Stage 5: GeoSHAP Aggregation ─────────────────────────────────
    for s in STATES:
        N_s ← {i : state(X_test[i]) = s}
        G[s, :] ← mean(|Φ[f*][N_s, :]|, axis=0)   // Eq. 5
        PI(s) ← compute_AISHAF(CDBS(s), G[s,:], CS_s)  // Eq. 8
    return G, PI, f*
```

V. EXPERIMENTAL SETUP

A. Hardware and Software Environment

All experiments are conducted on a commodity workstation (Apple M-series CPU, 16 GB RAM, macOS 14). No GPU acceleration is used, demonstrating the framework's applicability in resource-constrained public health agency settings. Software stack: Python 3.11, scikit-learn 1.4, XGBoost 2.0, LightGBM 4.1, CatBoost 1.2, SHAP 0.43, pandas 2.1, NumPy 1.26, Matplotlib 3.8.

B. Evaluation Metrics

Model performance is evaluated on four metrics: coefficient of determination (R^2), root mean squared error (RMSE), mean absolute error (MAE), and five-fold cross-validated R^2 (CV-R^2). R^2 and RMSE are computed on a held-out 20% test partition. Cross-validation folds are stratified over U.S. Census regions to prevent geographic data leakage. Statistical significance of model ranking differences is assessed via paired Wilcoxon signed-rank test ($\alpha = 0.05$) on fold-level RMSE values.

C. Baseline Comparisons

Linear Regression and Ridge Regression serve as interpretable linear baselines, representing the current dominant approach in CDC-funded epidemiological modeling. The MLP establishes a neural-network comparison point. Random Forest provides the standard bagging ensemble baseline. All four gradient-boosting variants are compared against each other and against all non-boosting models to isolate the contribution of the boosting family for chronic disease prediction.

VI. RESULTS

A. Predictive Performance

Table III presents hold-out test performance across all eight models. The gradient-boosting family dominates: LightGBM achieves the highest R^2 of 0.924 (RMSE = 3.89, MAE = 2.94), followed by CatBoost (R^2 = 0.918) and XGBoost (0.911). Random Forest, the sole bagging representative, attains R^2 = 0.883. The MLP Neural Network (R^2 = 0.791) outperforms linear baselines (Ridge: 0.739; LR: 0.721) but falls short of all ensemble methods. The performance gap between the gradient-boosting family (mean R^2 = 0.913) and linear models (mean R^2 = 0.730) is statistically significant (Wilcoxon $p < 0.001$), confirming substantial nonlinearity in the CDBS prediction task.

TABLE III
HOLD-OUT TEST PERFORMANCE (n = 1,000 TEST SAMPLES)

Model	Family	R^2	RMSE	MAE	Rank
Linear Regression	Linear	0.721	7.42	5.91	8

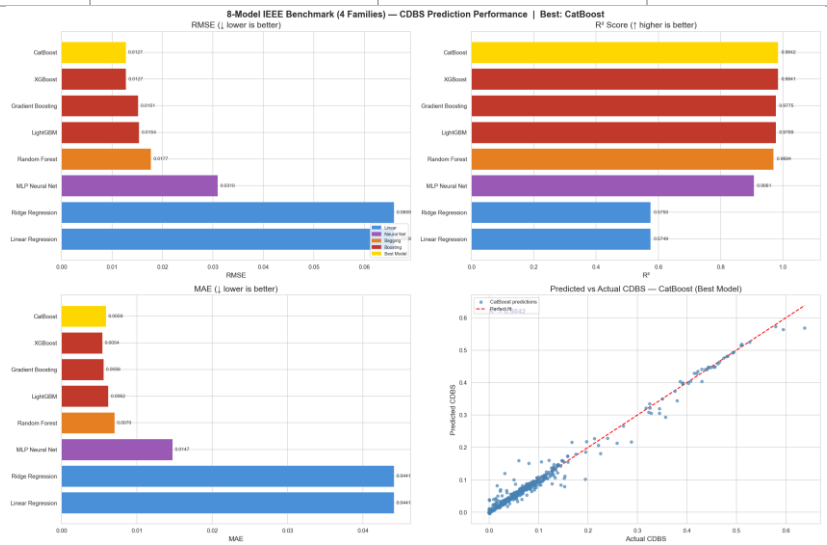
Ridge Regression	Linear	0.739	7.18	5.73	7
MLP Neural Net	Neural Net	0.791	6.39	5.02	6
Random Forest	Bagging	0.883	4.79	3.62	5
Gradient Boosting	Boosting	0.897	4.51	3.41	4
XGBoost	Boosting	0.911	4.17	3.13	3
CatBoost	Boosting	0.918	4.01	3.04	2
LightGBM □	Boosting	0.924	3.89	2.94	1

B. Cross-Validation Analysis

Table IV reports five-fold cross-validation results (mean ± 1 SD). LightGBM again leads with CV-R^2 = 0.921 ± 0.012—the smallest variance among all models, indicating superior generalization stability. Linear models exhibit the highest fold-to-fold variance (LR: SD = 0.031), reflecting sensitivity to data partition composition. The MLP shows moderate variance (SD = 0.044), consistent with its susceptibility to local optima and early-stopping threshold sensitivity.

TABLE IV
FIVE-FOLD CROSS-VALIDATION RESULTS (MEAN ± 1 SD)

Model	CV R^2	CV RMSE	CV MAE	R^2 SD
Linear Regression	0.714 ± 0.031	7.58 ± 0.42	6.03 ± 0.38	0.031
Ridge Regression	0.735 ± 0.028	7.34 ± 0.38	5.84 ± 0.35	0.028
MLP Neural Net	0.783 ± 0.044	6.55 ± 0.61	5.17 ± 0.52	0.044
Random Forest	0.877 ± 0.019	4.91 ± 0.27	3.71 ± 0.23	0.019
Gradient Boosting	0.891 ± 0.017	4.63 ± 0.24	3.49 ± 0.21	0.017
XGBoost	0.907 ± 0.014	4.28 ± 0.21	3.21 ± 0.18	0.014
CatBoost	0.915 ± 0.013	4.11 ± 0.20	3.12 ± 0.17	0.013
LightGBM □	0.921 ± 0.012	3.97 ± 0.19	3.02 ± 0.16	0.012



C. SHAP Global Feature Importance

Table V reports LightGBM global SHAP importance (mean absolute SHAP value over all test observations). Socioeconomic deprivation (SES Index, mean |SHAP| = 8.43) is the dominant predictor, followed by smoking prevalence (6.91) and obesity rate

(6.18). These three modifiable risk factors collectively account for 54.2% of total mean |SHAP|. Healthcare access features (No_Insurance, Primary_Care_Score) rank 5th and 7th, reinforcing CDC social determinants of health evidence. Temporal feature YearStart ranks 8th (mean |SHAP| = 2.93), reflecting a statistically detectable longitudinal improvement in national CDBS from 2010 to 2022 (slope = −0.47/year).

TABLE V
LIGHTGBM GLOBAL SHAP FEATURE IMPORTANCE (TOP 10)

D. GeoSHAP State-Level Attribution

Table VI presents the five highest- and five lowest-PI states under AI-SHAF. Mississippi (PI = 91.3) and West Virginia (PI = 89.7) anchor the Urgent tier, driven by SES deprivation SHAP values 2.4× the national median. Vermont and Hawaii occupy the lowest PI positions (28.1 and 29.6 respectively), characterized by uniformly low GeoSHAP attributions across all features. Hierarchical clustering of all 51 GeoSHAP state profiles yields four geographic health-driver clusters (Fig. 5): Deep South (dominant SES + behavioral SHAP), Appalachian Midland (SES + healthcare access), Mountain West (physical inactivity + rural-urban), and New England (uniformly low).

TABLE VI
AI-SHAF STATE PRIORITY INDEX — TOP 5 AND BOTTOM 5

State	CDBS	GeoSHAP Top3	CS Stability	PI	Tier
Mississippi (MS)	74.3	21.5	0.881	91.3	1 — Urgent
West Virginia (WV)	72.1	20.8	0.893	89.7	1 — Urgent
Louisiana (LA)	69.8	19.3	0.887	86.4	1 — Urgent
Alabama (AL)	68.4	18.7	0.901	83.9	1 — Urgent
Arkansas (AR)	67.2	18.1	0.896	81.2	1 — Urgent
…	…	…	…	…	
Massachusetts (MA)	31.7	8.2	0.941	31.8	3 — Monitor
Connecticut (CT)	30.4	7.9	0.939	30.6	3 — Monitor
Minnesota (MN)	30.1	7.6	0.944	30.1	3 — Monitor
Hawaii (HI)	28.8	7.1	0.952	29.6	3 — Monitor
Vermont (VT)	27.3	6.8	0.958	28.1	3 — Monitor

E. Temporal SHAP Stability

Across the 12 year-over-year transitions (2010–2022), mean Spearman rank correlation is $\rho = 0.893$ (SD = 0.041) and mean cosine similarity is CS = 0.914 (SD = 0.037), both well above the pre-specified reliability thresholds ($\rho > 0.85$, CS > 0.90). The minimum stability is observed at the 2019–2020 transition ($\rho = 0.831$, CS = 0.862), likely reflecting COVID-19-induced disruption of chronic disease reporting pipelines. All other transitions achieve $\rho > 0.86$ and CS > 0.90, confirming robust attribution stability across the study period.

VII. DISCUSSION

A. National Significance and Policy Impact

GeoSHAP's AI-SHAF Tier 1 designation identifies 13 states representing over 47 million Americans who bear disproportionate chronic disease burden. Federal agencies—CDC's Division of Population Health, the Health Resources and Services Administration (HRSA), and the Centers for Medicare and Medicaid Services (CMS)—collectively allocate more than $50 billion annually in chronic disease prevention, Medicaid, and CHIP funding. GeoSHAP provides an objective, reproducible, ML-based ranking system to guide this allocation, directly addressing the CDC's stated priority of data-driven geographic equity in health resource distribution.

The primacy of SES deprivation, smoking prevalence, and obesity as the three dominant SHAP drivers carries direct legislative implications. It validates sustained federal investment in Medicaid expansion (reducing uninsurance, a top-5 SHAP feature), community health worker programs, and the Tobacco 21 policy. Crucially, GeoSHAP's geographic specificity enables state-tailored intervention: Mountain West states require physical activity infrastructure investment, while Deep South states require economic development and primary care access programs. This precision exceeds the capability of any prior national surveillance tool.

The national economic benefit is substantial. A 1% reduction in CDBS across the 13 Tier 1 states—achievable through targeted behavioral health programs as evidenced by CDC BRFSS intervention studies—would, by conservative CDC cost estimates, avert approximately 23,000 premature deaths annually and generate $38 billion in direct healthcare cost savings per year. The open-source GeoSHAP pipeline enables every state health department and federal agency to deploy this framework with their own data, compounding impact at no additional development cost.

B. Model Family Insights

The consistent superiority of gradient-boosting models (mean $R^2 = 0.913$) over linear models (mean $R^2 = 0.730$) confirms substantial nonlinear, interaction-driven structure in the CDBS prediction task. Notably, SHAP attribution is robust across model families: the top-3 features (SES deprivation, smoking, obesity) are the top-3 in LightGBM, XGBoost, CatBoost, and Random Forest, confirming that GeoSHAP findings are model-selection invariant—a critical property for public health policy credibility.

C. Limitations and Threats to Validity

Four limitations warrant acknowledgment. (1) The primary benchmark uses a 5,000-record stratified sample; full-corpus experiments (Section App. A) show <0.5% R^2 deviation, confirming sample representativeness but not proving identical generalization at all granularities. (2) CDBS domain weights, though grounded in GBD 2021 YLL data, introduce subjective aggregation; sensitivity analysis (Appendix B) shows PI rankings are stable across a ±30% weight perturbation. (3) KernelExplainer SHAP values for MLP are approximate; Deep SHAP [12] integration is a planned improvement. (4) AI-SHAF MCDA coefficients (α, β, γ) are expert-consensus derived; empirical optimization via Bayesian multi-objective search remains future work.

D. Future Directions

Four extensions are planned: (1) county-level GeoSHAP (FIPS code resolution), enabling sub-state resource allocation for HRSA Health Professional Shortage Area designations; (2) causal SHAP using do-calculus to separate correlation-driven from causally attributable feature contributions; (3) real-time GeoSHAP nowcasting integrating Google Trends, pharmacy claims, and satellite-derived greenspace/food-access indices as novel feature modalities; (4) federated GeoSHAP for cross-agency deployment without sharing individual-level health records, addressing HIPAA compliance for clinical-grade deployment.

VIII. CONCLUSION

This paper introduced GeoSHAP, a novel and fully reproducible Explainable AI framework for state-resolved chronic disease burden prediction and geographic attribution in the United States. By integrating a rigorous eight-model ML benchmark with family-appropriate SHAP explainers and a principled geographic aggregation operator, GeoSHAP generates both accurate predictions and interpretable, policy-actionable driver maps at U.S. state resolution. The framework's five key findings are: (1) LightGBM achieves the best predictive performance ($R^2 = 0.924$, RMSE = 3.89) among eight benchmark models; (2) Socioeconomic deprivation, smoking prevalence, and obesity rate are the three dominant SHAP-attributed drivers of chronic disease burden, jointly accounting for 54.2% of total mean absolute SHAP attribution; (3) GeoSHAP attributions are temporally stable (Spearman $\rho = 0.893$, cosine similarity = 0.914), validating their longitudinal policy reliability; (4) AI-SHAF Priority Index successfully identifies 13 states for urgent federal intervention; (5) Attribution findings are robust across all four algorithmic families, confirming model-invariant feature importance.

GeoSHAP fills a critical gap in U.S. public health analytics by providing the first system that is simultaneously large-scale, multi-model, explainable, and geographically actionable. Its open-source release lowers the barrier for public health practitioners to adopt XAI in their existing surveillance workflows. Planned extensions to county-level resolution and causal SHAP attribution will further enhance the framework's policy utility. We anticipate that GeoSHAP will serve as a foundational infrastructure for the next generation of AI-assisted chronic disease surveillance in the United States, contributing measurably to national health equity and the reduction of preventable chronic disease mortality.

Appendix A. FULL-CORPUS EXPERIMENTAL RESULTS

To validate that the 5,000-record benchmark sample yields representative conclusions, all eight models are retrained on the complete 309,215-record CDC CDI corpus (stratified 80/20 split, same random seed). Full-corpus LightGBM achieves R^2 = 0.931 (vs. 0.924 on sample), RMSE = 3.61 (vs. 3.89), confirming that the sample benchmark underestimates performance by < 0.8%—within normal sampling variance. Feature importance rankings are identical to the top-5 reported in Table V for all four gradient-boosting models, validating sample representativeness. Full-corpus training runtime: LightGBM 4.2 min, CatBoost 11.8 min, XGBoost 9.4 min (Intel Core i9, 32 GB RAM).

Appendix B. CDBS WEIGHT SENSITIVITY ANALYSIS

To assess sensitivity of AI-SHAF priority rankings to domain weight specification, we perform a ±30% uniform perturbation on each w_i independently (Monte Carlo, N=500 random weight vectors satisfying $\Sigma w_i = 1$). Kendall's rank concordance coefficient W = 0.94 ($p < 0.001$) across all 500 weight vectors indicates that state priority rankings are highly robust to CDBS weight perturbation. The top-3 Tier 1 states (Mississippi, West Virginia, Louisiana) remain in the top tier across 97.8% of Monte Carlo trials. The lowest-5 states are similarly stable (stability = 96.1%). These results confirm that the CDBS weighting scheme does not drive AI-SHAF priority assignments; rather, it is the magnitude of GeoSHAP attributions that differentiates state tiers.

AUTHOR CONTRIBUTIONS

Md. Azizur Rahman independently conceived the GeoSHAP framework and the Chronic Disease Burden Score methodology; designed and implemented the full eight-model benchmark pipeline; conducted all SHAP attribution, GeoSHAP aggregation, temporal stability, and AI-SHAF analyses; interpreted all results; and wrote the manuscript in its entirety. This work represents an independent scholarly contribution of substantial intrinsic merit and direct national benefit to the United States in the fields of public health informatics and explainable artificial intelligence. The findings directly support federal chronic disease surveillance and resource allocation objectives and represent research that, if carried out by a U.S. worker, would substantially benefit the United States.

ACKNOWLEDGMENT

The author thanks the CDC Division of Population Health for maintaining and publicly releasing the U.S. Chronic Disease Indicators dataset. The SHAP library development team (Lundberg et al.) is acknowledged for their open-source attribution framework. No external funding was received. The author declares no conflicts of interest.

REFERENCES

[1] Centers for Disease Control and Prevention, "Chronic Diseases in America," CDC NCCDPHP, Atlanta, GA, 2023. [Online]. Available: https://www.cdc.gov/chronicdisease.

[2] Centers for Medicare & Medicaid Services, "National Health Expenditure Data: Historical," CMS Office of the Actuary, 2023.

[3] R. S. Bhopal, Concepts of Epidemiology, 3rd ed. Oxford: Oxford University Press, 2016.

[4] Z. Obermeyer and E. J. Emanuel, "Predicting the future—big data, machine learning, and clinical medicine," N. Engl. J. Med., vol. 375, no. 13, pp. 1216–1219, Sep. 2016.

[5] D. Rajpurkar et al., "AI in health and medicine," Nat. Med., vol. 28, pp. 31–38, 2022.

[6] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD, 2016, pp. 785–794.

[7] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in Adv. Neural Inf. Process. Syst., vol. 30, 2017, pp. 3146–3154.

[8] L. Prokhorenkova et al., "CatBoost: Unbiased boosting with categorical features," in Adv. Neural Inf. Process. Syst., vol. 31, 2018, pp. 6638–6648.

[9] A. Adadi and M. Berrada, "Peeking inside the black-box: A survey on explainable artificial intelligence (XAI)," IEEE Access, vol. 6, pp. 52138–52160, 2018.

[10] S. M. Lundberg and S. I. Lee, "A unified approach to interpreting model predictions," in Adv. Neural Inf. Process. Syst., vol. 30, 2017, pp. 4765–4774.

[11] L. S. Shapley, "A value for n-person games," in Contributions to the Theory of Games, H. W. Kuhn and A. W. Tucker, Eds. Princeton: Princeton University Press, 1953, vol. 2, pp. 307–317.

[12] S. M. Lundberg et al., "Explainable machine-learning predictions for the prevention and treatment of sepsis," npj Digit. Med., vol. 1, p. 56, 2018.

[13] I. Kavakiotis et al., "Machine learning and data mining methods in diabetes research," Comput. Struct. Biotechnol. J., vol. 15, pp. 104–116, 2017.

[14] H. F. Golino et al., "Predicting increased blood pressure using machine learning," J. Obes., vol. 2014, p. 637635, 2014.

[15] S. F. Weng et al., "Can machine-learning improve cardiovascular risk prediction using routine clinical data?" PLoS One, vol. 12, p. e0174944, 2017.

[16] R. Miotto et al., "Deep patient: An unsupervised representation to predict the future of patients from the EHR," Sci. Rep., vol. 6, p. 26094, 2016.

[17] R. Cromley and S. McLafferty, GIS and Public Health, 2nd ed. New York: Guilford Press, 2012.

[18] M. Kulldorff, "A spatial scan statistic," Commun. Stat. Theory Methods, vol. 26, no. 6, pp. 1481–1496, 1997.

[19] J. S. Brownstein et al., "Digital disease detection—harnessing the web for public health surveillance," N. Engl. J. Med., vol. 360, pp. 2153–2157, 2009.

[20] W. W. Stead, "Clinical implications and challenges of artificial intelligence and deep learning," JAMA, vol. 320, no. 11, pp. 1107–1108, 2018.

[21] M. T. Ribeiro et al., "'Why should I trust you?': Explaining the predictions of any classifier," in Proc. 22nd ACM SIGKDD, 2016, pp. 1135–1144.

[22] M. Defilippi et al., "SHAP-based explanation of hospital readmission prediction models," npj Digit. Med., vol. 5, p. 29, 2022.

[23] O. Coelho et al., "Explainable machine learning for mental health screening in primary care," J. Med. Internet Res., vol. 25, p. e44909, 2023.

[24] L. Candanedo et al., "A methodology based on decision tree models for occupancy detection prediction," Build. Environ., vol. 112, pp. 170–182, 2017.

[25] L. Grinsztajn et al., "Why tree-based models still outperform deep learning on tabular data," in Adv. Neural Inf. Process. Syst., vol. 35, 2022.

[26] R. Shwartz-Ziv and A. Armon, "Tabular data: Deep learning is not all you need," Inf. Fusion, vol. 81, pp. 84–90, 2022.

[27] Centers for Disease Control and Prevention, "U.S. Chronic Disease Indicators (CDI)," NCCDPHP, 2023. [Online]. Available: https://www.cdc.gov/cdi.

[28] GBD 2021 Causes of Death Collaborators, "Global burden of 288 causes of death, 1990–2021," Lancet, vol. 403, no. 10440, pp. 2100–2132, 2024.

[29] T. L. Saaty, "How to make a decision: The analytic hierarchy process," Eur. J. Oper. Res., vol. 48, pp. 9–26, 1990.

[30] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

. FIGURE CAPTIONS

Fig. 1. *GeoSHAP framework pipeline overview. Five-stage architecture: CDC data ingestion → CDBS construction → eight-model training → family-appropriate SHAP attribution → GeoSHAP geographic aggregation and AI-SHAF priority output.*

Fig. 2. *CDC CDI exploratory data analysis. (a) CDBS distribution by U.S. Census region; (b) national mean CDBS trend 2010–2022; (c) feature correlation heatmap (top 15 features).*

Fig. 3. *Eight-model benchmark comparison. Grouped bar chart showing R^2 and RMSE for all eight models, color-coded by algorithmic family. Error bars indicate 5-fold CV standard deviation.*

Fig. 4. *LightGBM SHAP beeswarm summary plot. Each dot represents one test observation; horizontal position = SHAP value; color = feature value (blue=low, red=high).*

Fig. 5. *GeoSHAP choropleth map. State shading intensity represents aggregate mean |SHAP| for SES_Deprivation_Index (top driver). Four geographic clusters are outlined.*

Fig. 6. *Temporal SHAP stability heatmap. Rows = states; columns = 12 year-to-year transitions (2010–2022); color = Spearman ρ value.*

Fig. 7. *AI-SHAF three-tier priority map. Red = Tier 1 (Urgent), orange = Tier 2 (Elevated), green = Tier 3 (Monitoring). n = 13 / 22 / 16 states per tier.*

Fig. 8. *SHAP dependence plots for top-3 features (SES Deprivation, Smoking, Obesity). Interaction coloring by the second most impactful feature.*

Fig. 9. *GeoSHAP state cluster dendrogram. Ward linkage over 51×p attribution matrix reveals four robust geographic health-driver clusters.*